

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Narappagari Sravya	Reg. No.:	192472192
Section / Batch:	B.E CSE AI / 2024	Date:	31-07-2036

Code of Conduct

I Narappagari Sravya(192472192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: sravya

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1.Aim

To analyze the words by separating the root and suffix, identify whether the suffix is inflectional or derivational, and normalize them to a common base form.

Input Word	Root	Suffix	Type	Morphological Breakdown	Normalized Output
connected	connect	-ed	Inflectional (Past tense)	connect + ed	connect
connecting	connect	-ing	Inflectional (Present participle)	connect + ing	connect
connection	connect	-ion	Derivational (Noun formation)	connect + ion	connect

Result

All words are normalized to the base form **connect** for efficient document retrieval.

2. Morphological Parsing – unhappy, happiness, happily

Aim

To identify prefixes, suffixes, and root words using rule-based parsing for sentiment analysis.

Input Word	Prefix	Root	Suffix	Type	Morphological Breakdown
unhappy	un-	happy	—	Derivational	un + happy
happiness	—	happy	-ness	Derivational	happy + ness
happily	—	happy	-ly	Derivational	happy + ly

Result

The common base form is **happy**, maintaining semantic consistency for sentiment analysis.

3. Stemming – played, player, playing

Aim

To remove prefixes/suffixes and obtain a common stem for search engine preprocessing.

Input Word	Stem	Affix Removed	Type	Morphological Breakdown
played	play	-ed	Inflectional	play + ed
player	play	-er	Derivational	play + er
playing	play	-ing	Inflectional	play + ing

Result

The extracted common stem is **play**, ensuring consistent indexing.

4. Finite-State Morphological Parsing – writes, writing, written

Aim

To represent word formation using finite-state transitions and normalize all forms to the common root.

Input Word	State Transition	Morphological Breakdown	Pattern	Normalized Output
writes	write → +s	write + s	Regular Inflection	write
writing	write → remove e → +ing	write → writing	Regular Inflection	write
written	write → irregular → written	write → written	Irregular Inflection	write

Finite-State Transitions

- **writes:** Start → **write** → add **-s** → writes
- **writing:** Start → **write** → remove **e** → add **-ing** → writing

- **written:** Start → **write** → irregular transition → written

Result

All forms are mapped to the common root **write**.

5. Porter Stemmer – relational, relation, relate

Aim

To apply Porter stemming rules and obtain a common stem for document retrieval.

Input Word	Stemming Steps	Intermediate Form	Final Stem
relational	relational → remove -ational → relate → remove -e	relat	relat
relation	relation → remove -ion	relat	relat
relate	relate → remove -e	relat	relat

Result

All words are reduced to the common stem **relat**, which is the output produced by the Porter Stemmer.